1 Rational Expressions

The quotient of two polynomials is called a **rational expression**, where the denominator is never zero.

Examples include $5/6$, $\frac{3x+8}{2x-9}$, $6x+5$

The last one can be written as $\frac{6x+5}{1}$ which fits the definition.

1.1 Simplifying Rational Expressions

1. Completely factor numerators and denominators. 2. Reduce common factors.

1.2 Multiplying Rational Expressions

1. Completely factor all numerators and denominators.
2. Reduce all common factors.
3. Either multiply the denominators, using FOIL if needed, or leave it in factored form.

1.3 Dividing Rational Expressions

Division of fractions involves multiplying by the *reciprocal* of the divisor.

1.4 Adding and Subtracting Rational Expressions

If they have the same denominator then add/subtract as usual making sure to simplify at the end.
If they *dont* have the same denominator then you need to find the Lowest Common Denominator (LCD).

1.5 Complex Fractions

If the numerator or denominator or both contain fractions the expression is called a **complex fraction.**

1. Simplify both numerator and denominator expressions into single fraction expressions.
2. Divide the numerator by the denominator and simplify if possible.

1.6 Proportion, Direct Variation, Inverse Variation, Joint Variation

Proportion is an equation stating that two rational expressions are equal. Simple ones are solved with the *Cross Product Rule*

> **Cross Product Rule**
> if $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$.

Direct Variation is when 'y varies directly as x' or 'y is directly proportional to x'. This may be inteperted in two ways:

1. $\frac{y}{x} = k$

For some constant k - k is called the **constant of proportionality**. This is used when the constant is the desired result.

2. $\frac{y_1}{x_1} = \frac{y_2}{x_2}$

This is when the desired result is either an original or new value of x or y.

Inverse Variation is when 'y varies inversely (opposite) to x'.

Joint Variation is when one variable varies as the product of the other variables. The phrase goes 'y varies jointly as x and z':

1. $\frac{y}{xz} = k$

if the constant is desired.

2. $\frac{y_1}{x_1 z_1} = \frac{y_2}{x_2 z_2}$

if one of the variables is desired.